

Is the ceiling the model's or the game's?

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design written 2026-10-08; not yet agreed; no run started

1 Question

The local model cannot improve the 17-point reference bot: 0 of 30 single revisions, and two 40-generation chains never left its behaviour (`findings/2026-10-07-local-model-ceiling.md`). Every accumulation experiment is bounded above by the innovation rate at the level the population has reached, so before any population run on a new model we need its rate at three levels. This is a measurement, not a hypothesis test, but it has a decision attached: whether to run populations on this model at all, and whether the revise prompt is the bottleneck.

2 Setting

Three incumbents: the weak student (held-out 2.75), the 16-point bot (IGGI, 15.8), the 17-point bot (Piers, 17.0). One revision each, nothing delivered, feedback traces varying across 30 replicates (experiment seeds 400–429, as in the local probe), verification off and no adoption, held-out paired evaluation on 300 deals and a common 300-deal set so levels are comparable. Hosted model, temperature 0, seed fixed (providers may not honour it; replicates are seeds, not replays).

3 What we measure

For incumbent level ℓ , the innovation rate $\epsilon(\ell) = \mathbb{P}(\Delta > 0.5)$ with Δ the paired held-out improvement, Beta(1,1) posterior and 95% interval; the gain distribution given a gain; the no-op rate (candidate byte-identical to the parent); the inadmissible rate (fails the sandbox or the smoke game). The no-op rate is the diagnostic for “the prompt tells the model to leave good code alone” versus “the model cannot find an improvement”.

4 Prediction (2026-10-08)

For a flash-class open-weight model: $\epsilon(2.75) \geq 0.6$ with gains around 8; $\epsilon(15.8) \in [0.1, 0.3]$; $\epsilon(17.0) \in [0.05, 0.2]$ with gains of 0.5 to 2; no-op rate below 0.3. If $\epsilon(17.0)$ is again 0/30 and the no-op rate is above 0.5, the ceiling is in the prompt (it asks for a rewrite of the worst games' causes and the 17-point bot has few), and the next step is a prompt variant, not a bigger model.

5 Design

Three levels $\times$ 30 replicates, 90 calls, paired on experiment seeds with the local probe so the two models are compared on the same traces. Decision rule, fixed now: populations run on this model

only if the lower 95% bound of $\epsilon(17.0)$ exceeds 0.03; otherwise the prompt is revised and the probe re-run before any population spend. Figure: ϵ against level for the two models, with intervals; gain histograms.

6 Cost

90 calls, about one dollar at flash-class prices, twenty minutes in parallel. This is also the first paid probe of the provider: it reports price per call, latency and whether the provider is deterministic at temperature 0 with a seed.

7 Nearest prior work

Innovation-rate ladders for program-writing agents were not found; the closest are self-improvement curves in evolutionary code search (AlphaEvolve-style systems report best-so-far against iterations, not a rate by level). Phrase accordingly.